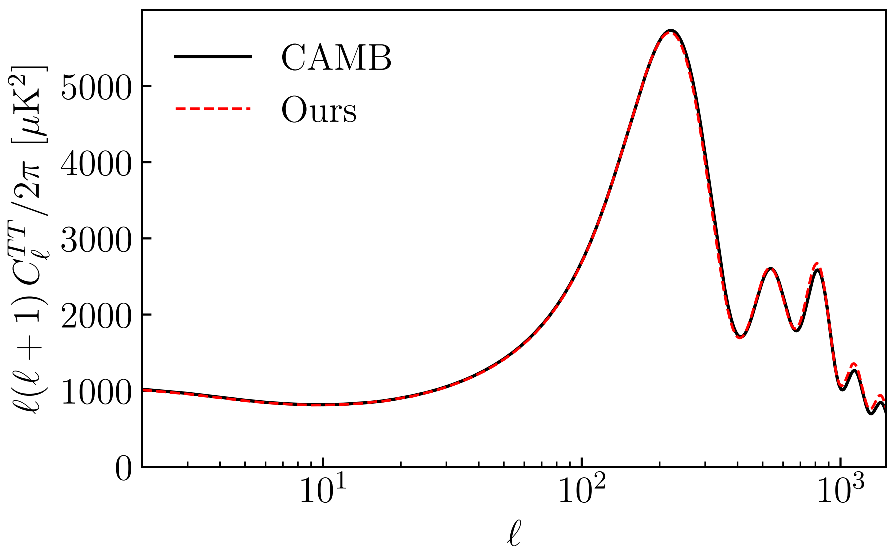
Boltzmann Code - CMB Angular Power Spectrum

In **perturbations.pdf** we built a Boltzmann solver that evolves CDM, baryons, photons, and massless neutrinos and did some fun things with it in **fun.pdf**. This week we'll use it to compute the CMB temperature angular power spectrum C_ℓ^{TT}.



Outline: The calculation is split into three pieces:

a. **Source function** $S_T(k,\eta)$: SW + Doppler + ISW

b. **Transfer function** $\Delta_\ell(k) = \int d\eta\, S_T(k,\eta)\, j_\ell(k\chi(\eta))$: Projecting the source onto multipoles.

c. **Angular spectrum** $C_\ell = 4\pi \int d\ln k\, \Delta_\ell^2(k)\, \Delta_\mathcal{R}^2(k)$: Integrates against the primordial curvature power $\Delta_\mathcal{R}^2(k) = A_s\,(k/k_\star)^{n_s-1}$.

Setup: Up until now we've being using some non-standard variables from the Bashinsky & Seljak (2003) paper. For computing the angular spectrum we'll convert back to to standard conformal Newton variables that are used in Baumann and in Ma & Bertschinger[1]. The metric potentials get swapped between the two conventions:

$$\boxed{\phi_N = \Psi^{\rm BS}, \qquad \psi_N = \Phi^{\rm BS}.} \tag{1}$$

For the matter variables (density, velocity, and higher-mulitpoles) we can convert to more standard variables with

$$\boxed{\begin{aligned} \delta_a^N &= (1+w_a)(d_a + 3\phi_N), \\ \theta_a^N &= k^2 u_a, \\ \sigma_a^N &= k^2 \pi_a, \\ F_{a,\ell}^N &= \tfrac{4}{3}\, k^\ell\, d_{\ell,a} \quad (\ell \geq 1), \\ F_{a,2}^N &= 2\sigma_a^N. \end{aligned}} \tag{2}$$

Part 1: Write a function `to_standard(y, k, a)` that returns a dictionary with entries `delta_g, theta_g, sigma_g, delta_n, theta_n, sigma_n, delta_c, theta_c, delta_b, theta_b, phi_N, psi_N` and higher multipoles using Eqs. (1)–(2).

Visibility function: Our old friend from `reionization.pdf`. The probability that a photon we observe today last-scattered at conformal time η is the **visibility function**

$$\boxed{g(\eta) = \tau'(\eta)\, e^{-\tau(\eta)}, \qquad \tau(\eta) = \int_\eta^{\eta_0} \tau'(\eta')\, d\eta',} \tag{3}$$

[1] We could continue the calculation in the non-standard variables but alas I didn't have time to work through things in terms of e.g. d_γ so we'll just fall-back on the canonical references.

where $\tau' = \sigma_T a n_e$ is the Thomson opacity. We will need g, its first derivative g', and $e^{-\tau}$ at a bunch of conformal times to compute the source function.

Part 2: Write a function `visibility_grid(etas)` that returns $(g, g', e^{-\tau})$ on an input conformal-time grid.

Source Function: The temperature source function can be written as

$$\boxed{S_T(k,\eta) = \underbrace{g\left(\frac{\delta_\gamma}{4} + \psi\right)}_{\text{SW}} + \underbrace{\frac{g'\,\theta_b + g\,\theta_b'}{k^2}}_{\text{Doppler}} + \underbrace{e^{-\tau}\,(\phi' + \psi')}_{\text{ISW}}} \tag{4}$$

Part 3: Write `los_source_grid(k, taus)` that evolves the perturbations with `solve_perturbations(k)`, samples the solutions on `taus`, applies `to_standard` at each time to get $(\delta_\gamma, \theta_b, \phi, \psi)$, and then computes $S_T(k,\eta)$ with Eq. (4).

Transfer function: Now that we have S_T we can compute the temperature transfer function

$$\boxed{\Delta_\ell(k) = \int_0^{\eta_0} d\eta\; S_T(k,\eta) j_\ell(k(\eta_0 - \eta)).} \tag{5}$$

Part 4: Write `delta_l_k(l_arr, k, S_T_grid, taus)` that returns $\Delta_\ell(k)$ for every ℓ in `l_arr` by trapezoidal integration of Eq. (5).

Sampling grids: Picking grids that are accurate without being too expensive is an art... We will do something inspired by CAMB's defaults:

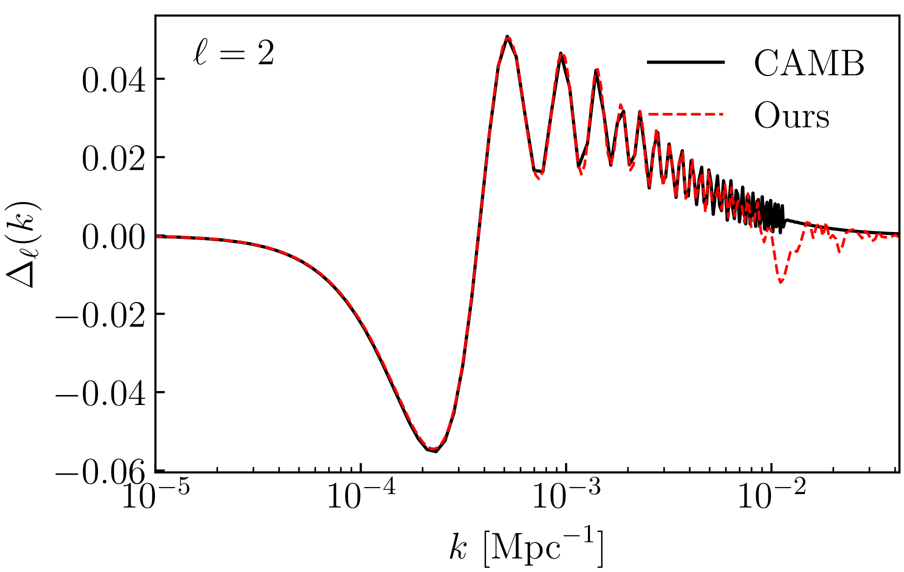
- **conformal-time grid** (`default_tau_grid`): a few points log-spaced before recombination ($\tau \lesssim 200$ Mpc), a hundred or so points densely sampling the visibility peak $\tau \in [200, 360]$ Mpc, and a few log-spaced points after.

- k-**grid** (`default_k_grids`): we need two of these.
  - Sparse `k_source` on which we evaluate S_T via `los_source_grid`. Source evaluation is expensive because each k requires solving the full perturbation system but the the source itself is smooth enough that we can get away with evalauting at a few k and interpolating.
  - Dense `k_dense` for the Bessel integral. j_ℓ does nothing but oscillate and we need to resolve these oscillations to get our integral to converge.

At this point you can test your code against the $\Delta_\ell(k)$ `CAMB` outputs.

```
ct = results.get_cmb_transfer_data('scalar')
camb_L   = ct.L            # the l-grid CAMB stored on
camb_q   = ct.q            # CAMB's k-grid
camb_Dlk = ct.delta_p_l_k[0, :, :]   # [source, l, k]; source=0 is T
```

What I get for $\ell = 2$ is the following. You should try this for different ℓ's.



Angular Spectrum: With $\Delta_\ell(k)$ on the dense k-grid, the angular spectrum follows from:

$$\boxed{C_\ell = 4\pi \int d\ln k \; \Delta_\ell^2(k)\; \Delta_{\mathcal{R}}^2(k), \qquad \Delta_{\mathcal{R}}^2(k) = A_s\left(\frac{k}{k_\star}\right)^{n_s-1}.} \tag{6}$$

Part 5: Write `cl_tt(taus, k_source, k_dense, l_arr)` that does the four steps end-to-end:

a. evaluate $S_T(k,\eta)$ on `k_source`

b. cubic-spline S_T in $\log k$ onto `k_dense`

c. integrate `delta_l_k` on `k_dense`

d. integrate $|\Delta_\ell|^2\Delta_{\mathcal{R}}^2$ in $\ln k$.

If everything goes well, you should be able to compare you outputted C_ℓ^{TT} to what CAMB outputs.

References:

- **Ma & Bertschinger (1995)**, *Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges*, astro-ph/9506072.
  The standard reference for linear perturbations.

- **Bashinsky & Seljak (2003)**, *Signatures of relativistic neutrinos in CMB anisotropy and matter clustering*, astro-ph/0310198.
  The paper whose notation we've been following.

- **Baumann**, *Cosmology*.
  Chapter 7 and Appendix B that you've covered in class.

- **Seljak & Zaldarriaga (1996)**, *A Line of Sight Approach to Cosmic Microwave Background Anisotropies*, astro-ph/9603033.